Title Page

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Running Course Analyzer

Analyzing Seoul's Best Running Courses
by Terrain, Weather & Difficulty

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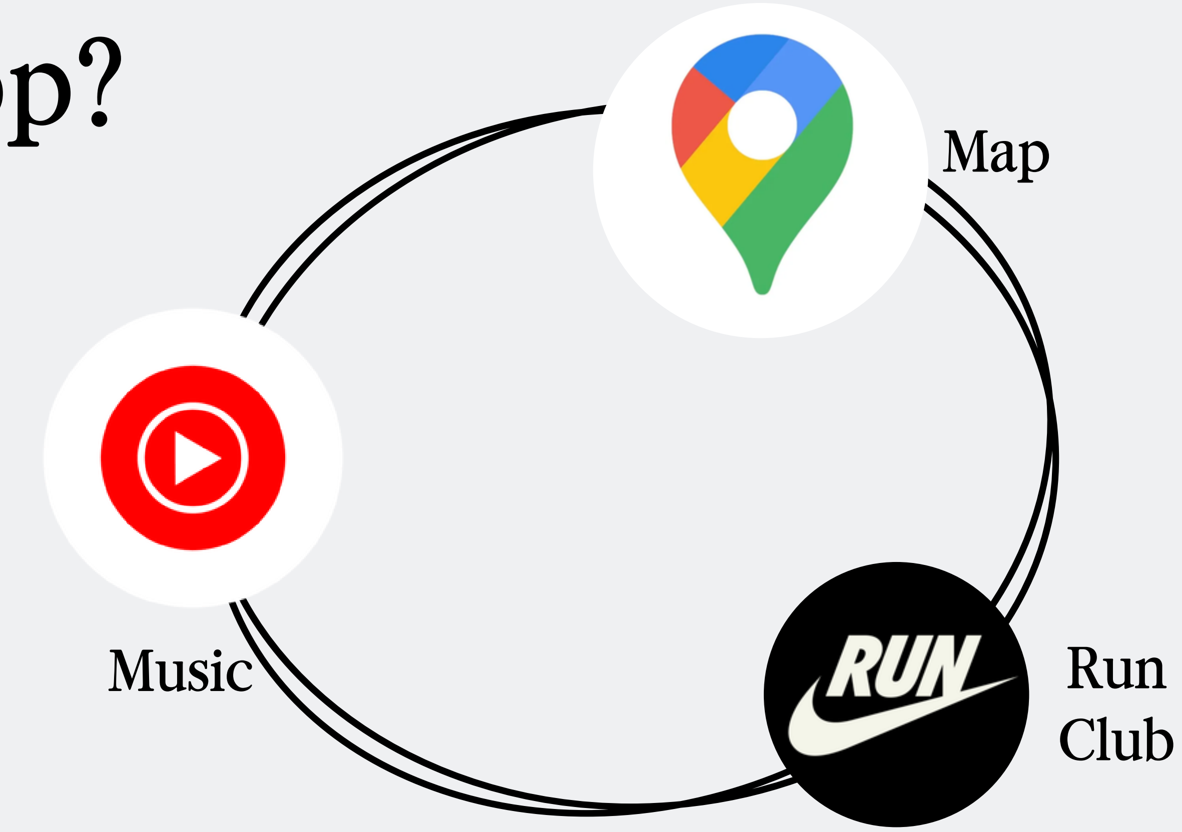
Why Did I Build This App?



Problem

Runners in Seoul must check a map for the route, a separate tool for elevation, and a weather app before heading out. I wanted to bring all of that into one screen.

Learning Goal

I wanted to build something I would actually use with real data, and see it work as a live application.

Challenge

Rather than analyzing a dataset handed to me, I wanted to go through the full process. Finding data, structuring it, and building something on top of it.

Title Page | Introduction

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What Powers the App?

Data Sources

- Course info (distance, surface, location)
  → Naver Map & Kakao Map
  → Seoul Open Data Plaza (data.seoul.go.kr)
- Elevation profiles
  → OpenStreetMap terrain data
  → Modeled per course (steep for Bukhansan vs. flat for Han River)
- Weather conditions
  → Simulated from Seoul seasonal averages (temperature, humidity, wind)

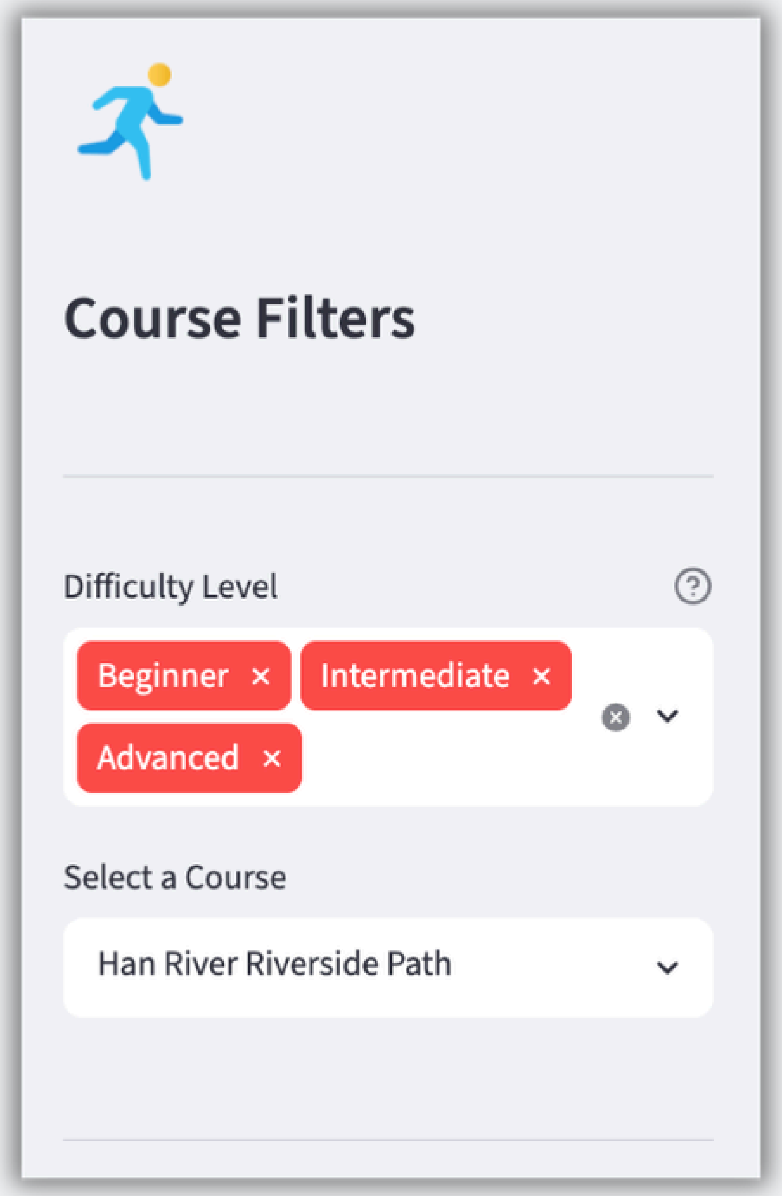
Tech Stack

- Streamlit 1.35 UI framework, sidebar, metrics
- Pandas 2.2 DataFrames, Styler tables
- NumPy 1.26 Elevation profile modeling
- Plotly 5.22 Interactive area & bar charts

What Can the App Do?

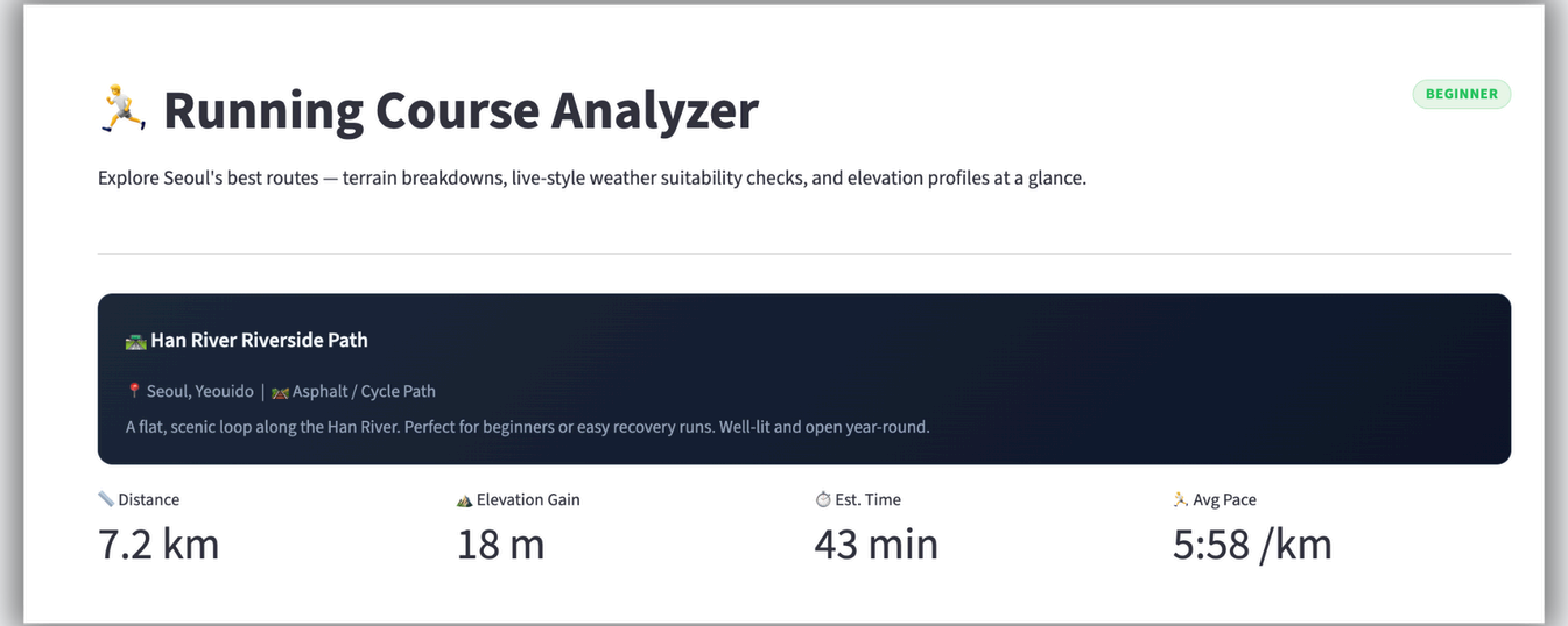
1 Course Filter & Selection

Sidebar multi-select for difficulty level. Selecting a course instantly refreshes the entire main panel, no reload needed.



What Can the App Do?

2 Course Overview & Key Metrics



Description card showing location and surface type.
Four metric boxes: Distance, Elevation Gain, Estimated Time, and Avg Pace.

Title Page Introduction App Demo

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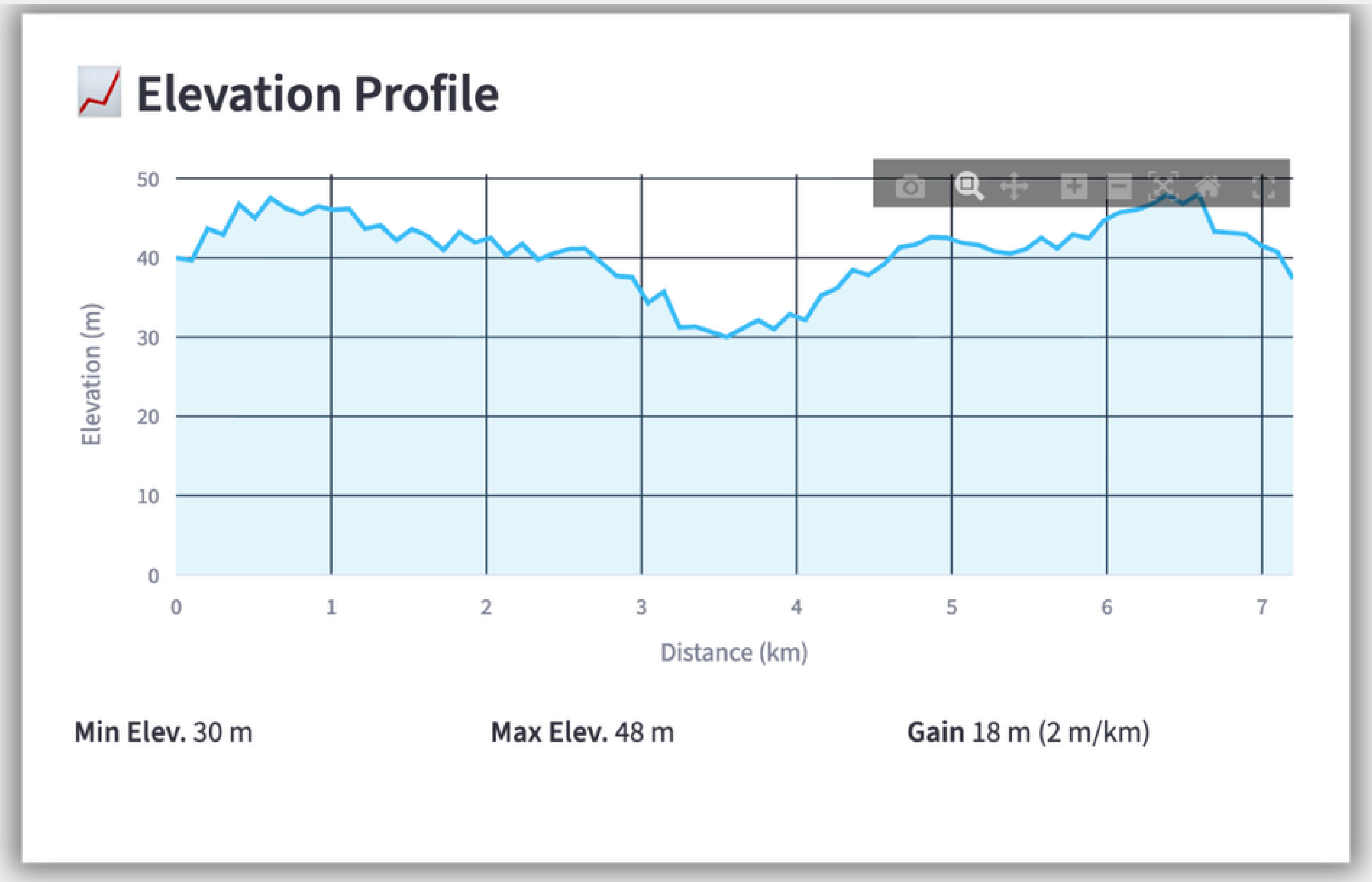
What Can the App Do?

3 Elevation Profile Chart

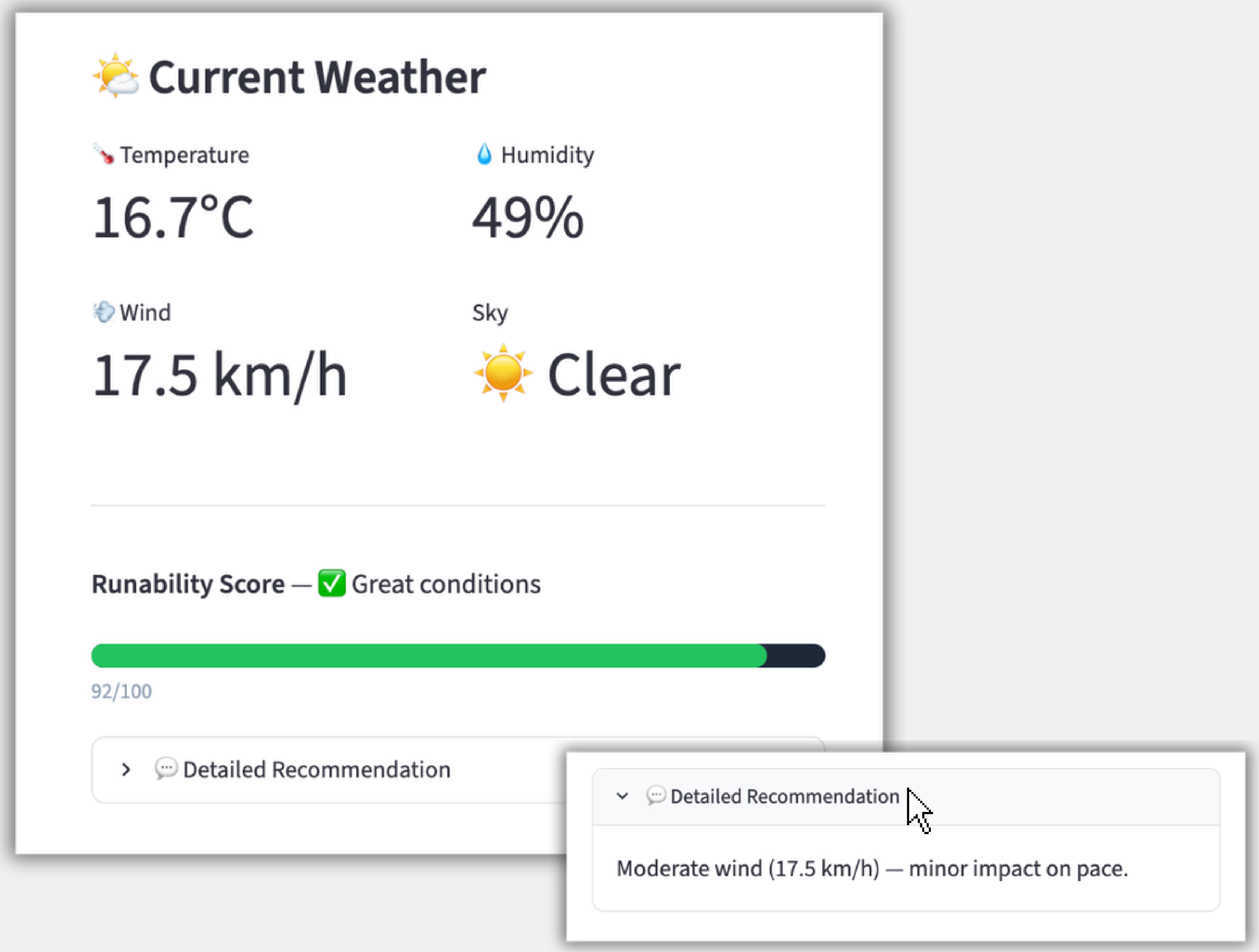


Interactive Plotly area chart. Hover to see exact elevation at any point.
Min / max elevation and gain per km shown below.

What Can the App Do?

4 Weather Suitability Score

Runability Score (0100) combining temp, humidity, wind & sky condition.
Bar turns green / yellow / red.
Course difficulty is factored in.



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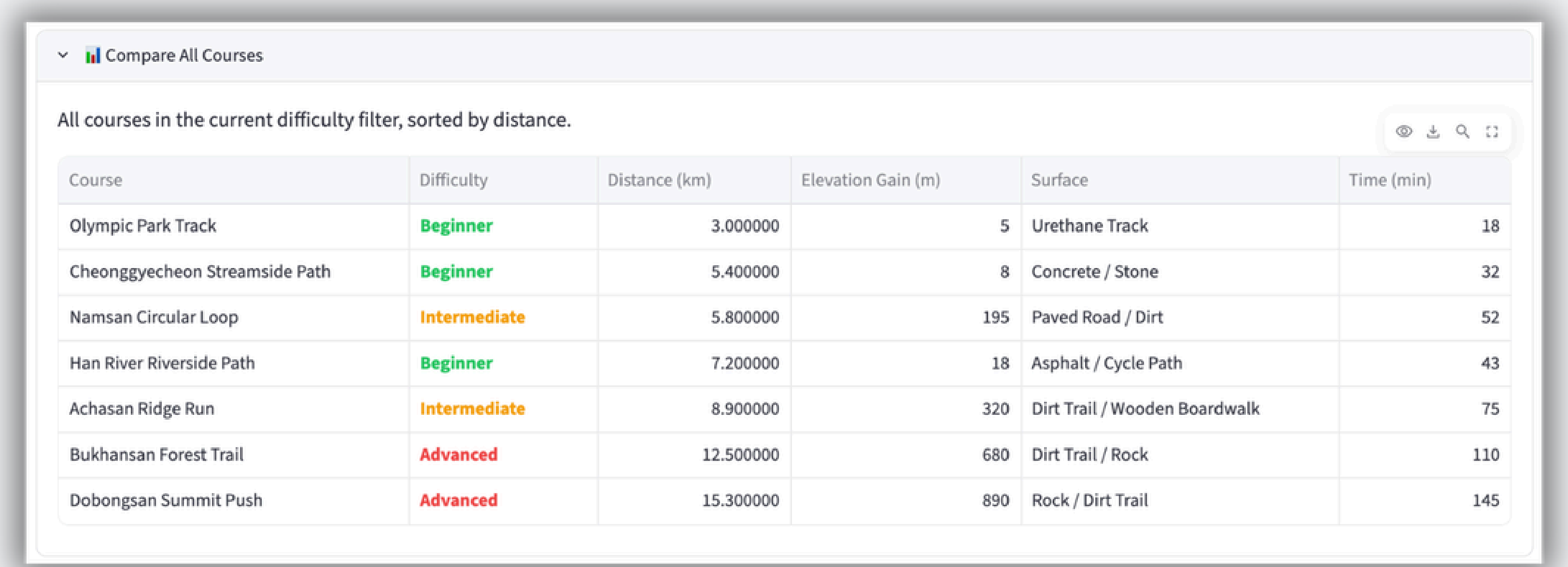
What Can the App Do?

5 Course Comparison Dashboard

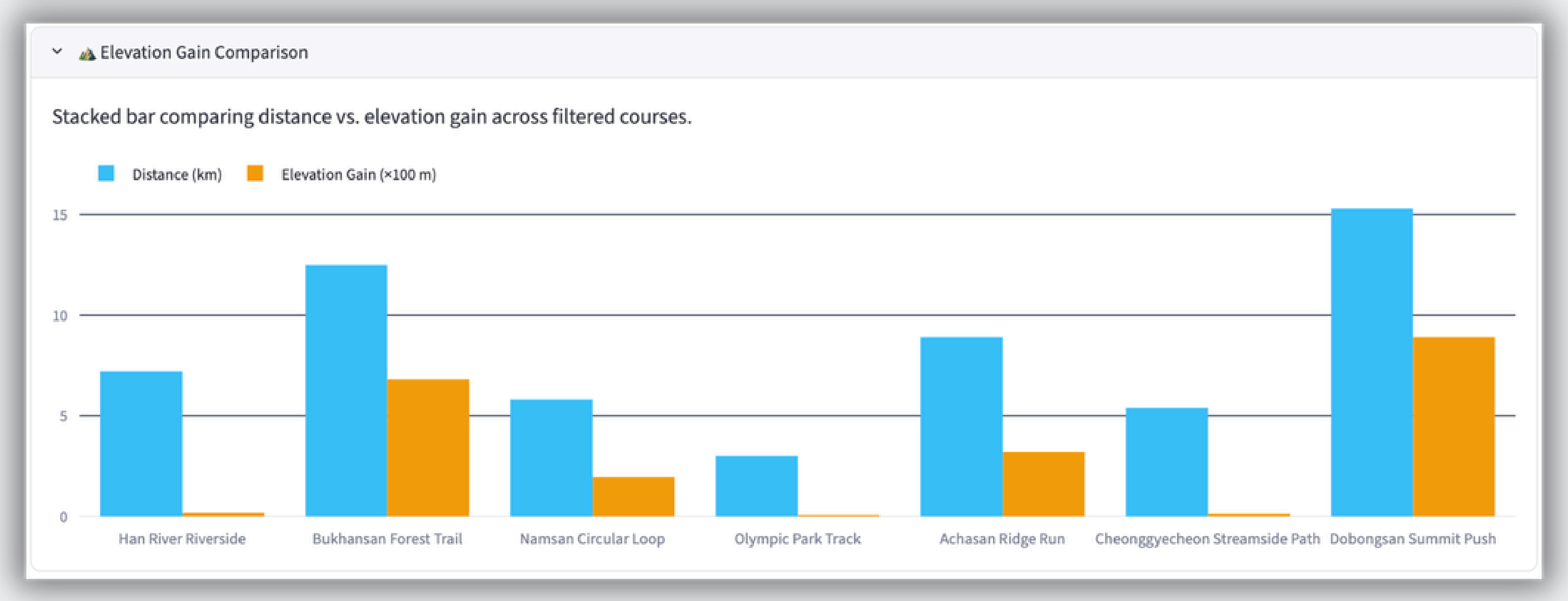
Expandable color-coded table + grouped bar chart comparing all filtered courses by distance and elevation gain side by side.



Compare All Courses

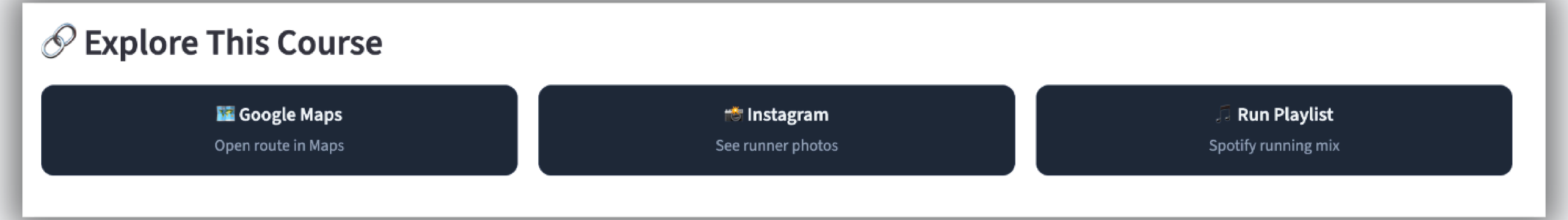
All courses in the current difficulty filter, sorted by distance.

Course	Difficulty	Distance (km)	Elevation Gain (m)	Surface	Time (min)
Olympic Park Track	Beginner	3.000000	5	Urethane Track	18
Cheonggyecheon Streamside Path	Beginner	5.400000	8	Concrete / Stone	32
Namsan Circular Loop	Intermediate	5.800000	195	Paved Road / Dirt	52
Han River Riverside Path	Beginner	7.200000	18	Asphalt / Cycle Path	43
Achasan Ridge Run	Intermediate	8.900000	320	Dirt Trail / Wooden Boardwalk	75
Bukhansan Forest Trail	Advanced	12.500000	680	Dirt Trail / Rock	110
Dobongsan Summit Push	Advanced	15.300000	890	Rock / Dirt Trail	145



What Can the App Do?

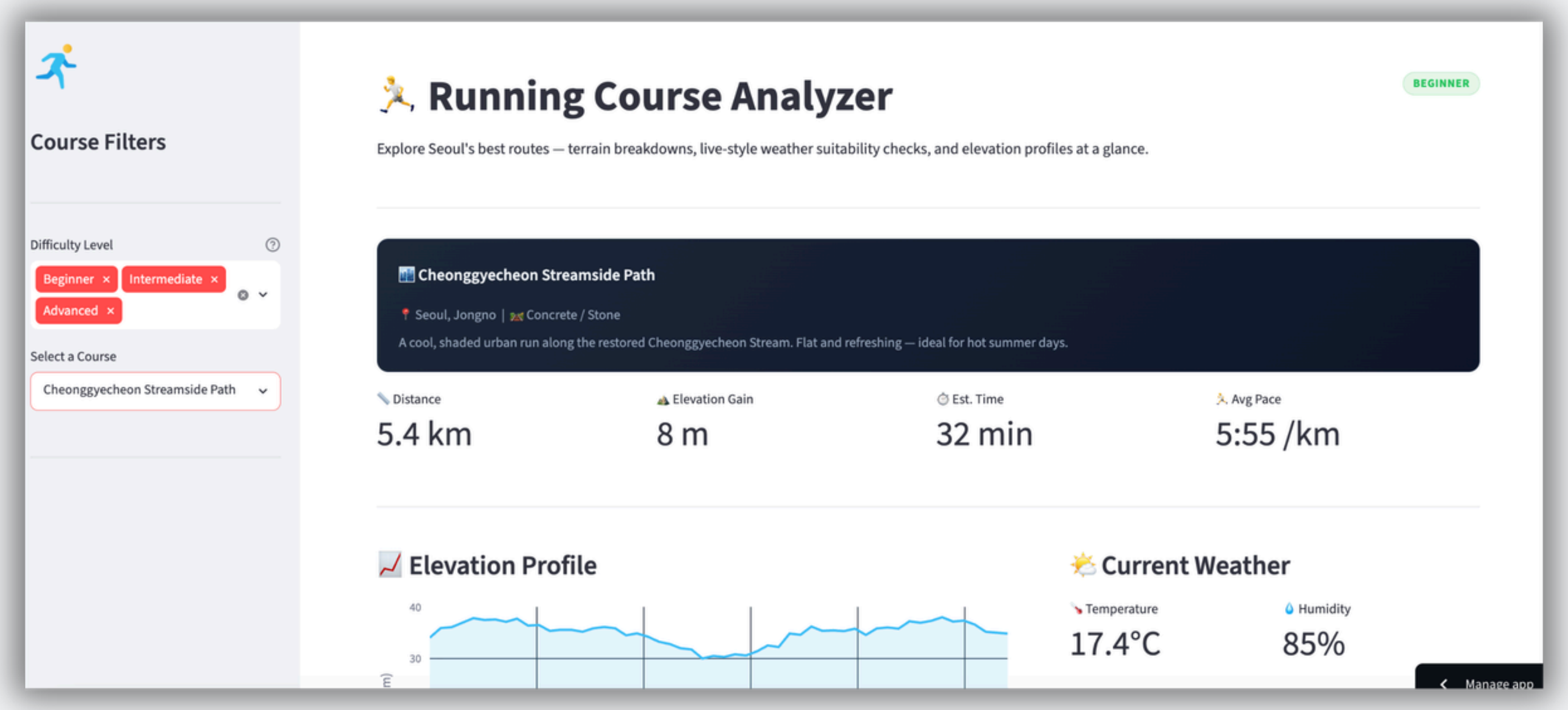
6 External Links : Explore & Go



Three one-tap buttons per course: open the route in Google Maps, browse runner photos on Instagram, and launch a Spotify playlist matched to the course's energy. Built to feel like an app you'd actually open before a run, not just a data table.

https://yeonhu-skkuappio-5tcxd6tmgadjtdfsbwunkj.streamlit.app/

What Did I Learn?



1 Designing for a real user changes everything.

If someone is about to go for a run right now, what do they need to see first?

2 Data only becomes useful in context.

A raw elevation number means very little on its own, presentation is part of the analysis.

3 Coding theory finally made sense in practice.

I've learned programming concepts across different courses : data structures, logic design, function architecture.

Title Page Introduction App Demo Conclusion

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Thank You

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Questions are welcome!